

# Kjartan van Driel

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## EDUCATION

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| <b>Iliad Intensive</b><br>Iliad, London<br>Selective month-long intensive in the mathematics of technical AI alignment: theory of RL, learning theory, interpretability, agent foundations, and scalable oversight.                                                                                                             | 2026           |
| <b>PhD Candidate</b><br>Complexity Science Hub & TU Wien, Vienna<br>ML methods for dynamic graph-structured data, applied to supply-chain networks to reveal vulnerabilities and support the green economic transition. Supervisor: Stefan Thurner.                                                                             | 2023 – present |
| <b>MSc Physics</b> (9.0/10 GPA; 4.0/4.0 US)<br>Universiteit van Amsterdam — Theoretical Physics track<br>Thesis: <i>Approximate Inference in Spiking Neural Networks</i> , Max Planck Institute for Dynamics and Self-Organization, Göttingen. Supervisors: V. Priesemann, F. Mikulasch, L. Rudelt.                             | 2020 – 2023    |
| <b>BSc Physics &amp; BSc Mathematics</b> ( <i>cum laude</i> )<br>Radboud Universiteit, Nijmegen<br>Thesis: <i>Path-Integral Control Theory Applied to Evolutionary Dynamics</i> , Donders Institute. Supervisors: H.J. Kappen, A. Nourmohammad.<br>Honours Programme: funded research at the University of Washington, Seattle. | 2017 – 2020    |

## PUBLICATIONS

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| <b>K. van Driel</b> , L.N. Ialongo, P.A. Astudillo-Estévez, S. Thurner. <i>Generalized Degrees for Scalable Discrete Time Dynamic Graph Generation</i> . Learning on Graphs Conference, 2025. <b>Oral</b> . <a href="#">[paper]</a> <a href="#">[code]</a> |
| <b>K. van Driel</b> , L. Rudelt, V. Priesemann, F.A. Mikulasch. <i>Prediction Mismatch Responses Arise as Corrections of a Predictive Spiking Code</i> . bioRxiv preprint, 2023. <a href="#">[paper]</a> <a href="#">[code]</a>                            |

## TALKS & CONFERENCES

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| London Geometry and Machine Learning Summer School (LOGML), London         | 2026 |
| European Forum Alpbach — <i>Complexity science in the green transition</i> | 2025 |
| Learning on Graphs Conference — Oral presentation                          | 2025 |
| Summer School on Economic Networks, University of Oxford                   | 2024 |
| ICLR — Volunteer                                                           | 2024 |
| CSH Winter School on Complex Systems                                       | 2024 |
| Summer School on Economic Complexity, Maastricht University                | 2024 |

## EXPERIENCE

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| <b>Quantitative Analyst</b> (part-time)<br>Gimli — Game-theoretic models for Flashbots MEV blind auction bid optimisation.              | 2021 – 2022 |
| <b>Intern Analyst</b> (part-time)<br>MindHash — Machine learning for object extraction and tracking in LiDAR point cloud data.          | 2020 – 2021 |
| <b>Blommers Coffee Roastery</b> , Nijmegen                                                                                              | 2020 – 2021 |
| <b>Undergraduate Researcher</b><br>University of Washington, Seattle — Computational research on constrained mutations in genome space. | 2020        |
| <b>Teaching Assistant / Tutor</b><br>Radboud Universiteit                                                                               | 2015 – 2020 |

## ACTIVITIES

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|---------------------------------------------------------------------|-------------|
| Reading group on EU industrial and economic policy (organiser), CSH | 2024        |
| Honours Programme, interdisciplinary research (Radboud)             | 2017 – 2020 |
| Symposium Committee, study association Marie Curie                  | 2017 – 2019 |
| Physics Bachelor Representative at high schools & universities      | 2017 – 2019 |

## SKILLS

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| <b>Programming</b> Python (PyTorch, PyTorch Geometric, JAX) · CUDA / Triton · C++ · $\LaTeX$ · Git · Linux / Slurm                                                                        |
| <b>Methods</b> Graph neural networks · Normalizing flows · Dynamic graph generation · Spiking neural networks · Variational inference · Network science · Probabilistic generative models |
| <b>Languages</b> Dutch (native) · English (fluent) · German (B2)                                                                                                                          |